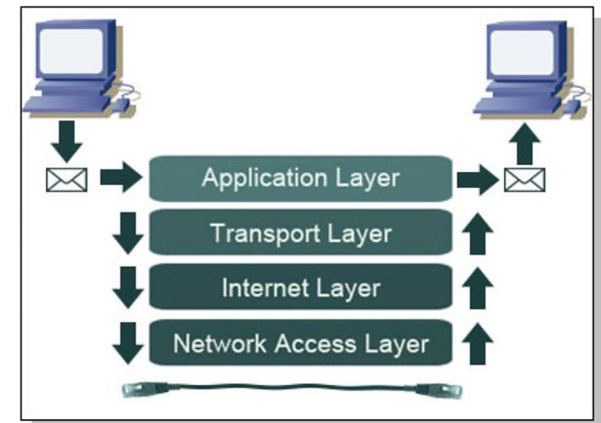
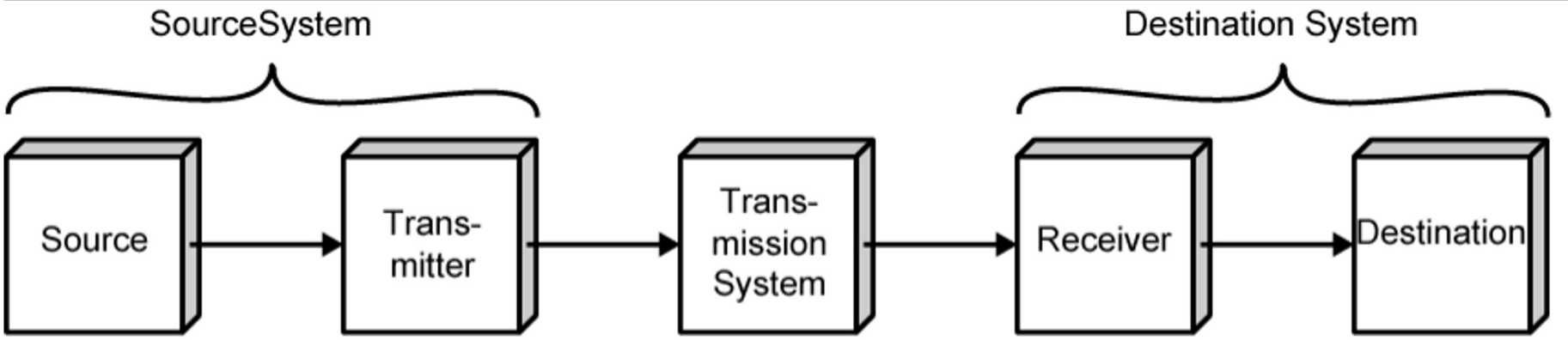


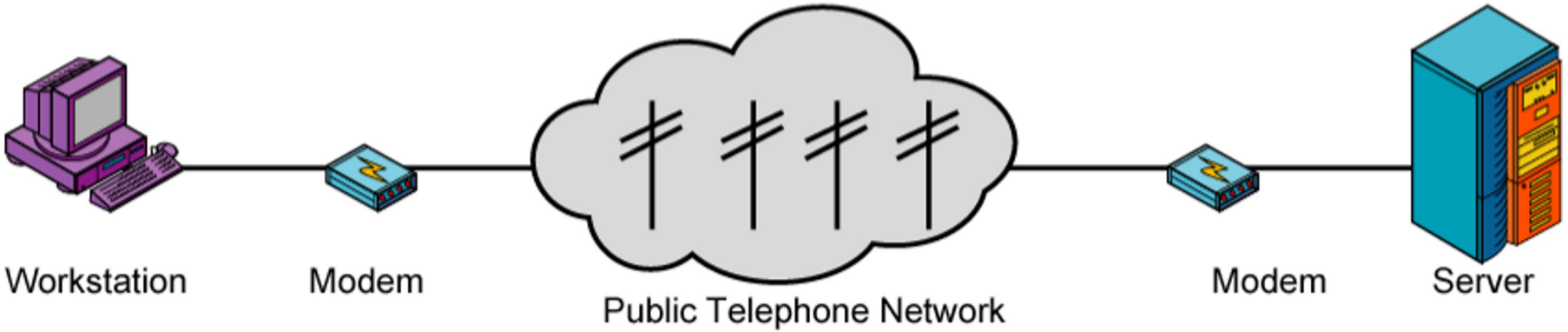
# Network, Internet and Internet of Things



# Communication Model

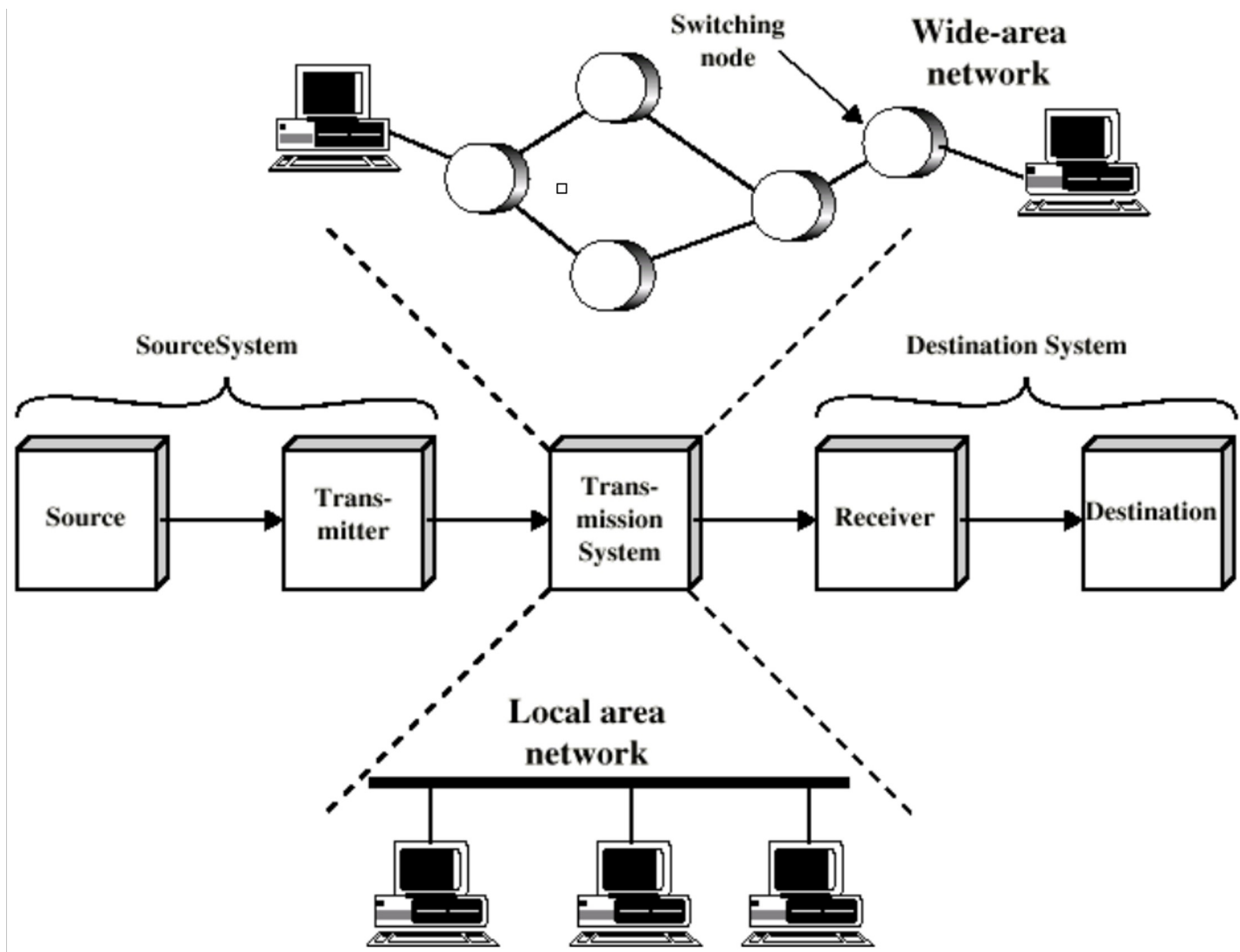


(a) General block diagram



(b) Example

# Conventional Communication Networks



# Local Area Networks (LANs)

- Characteristics

- Smaller scope
  - Building or small campus
- Usually owned by same organization as attached devices
- Data rates much higher

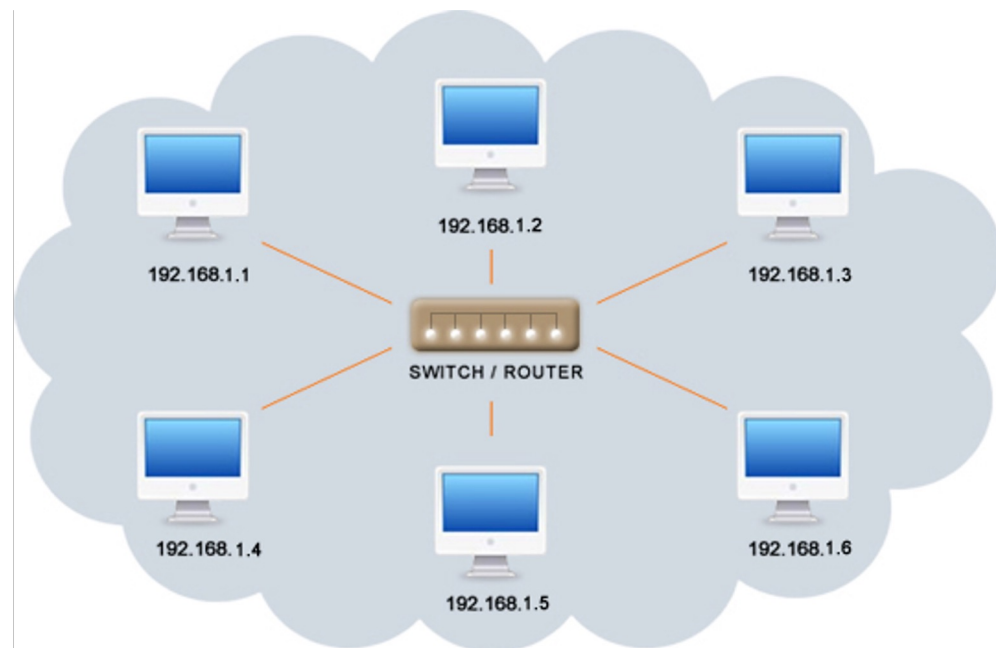
Token-Ring



- Categories

- Switched LANs
  - Ethernet
- Wireless LANs
- **ATM LANs**

**(Asynchronous Transfer Mode)**

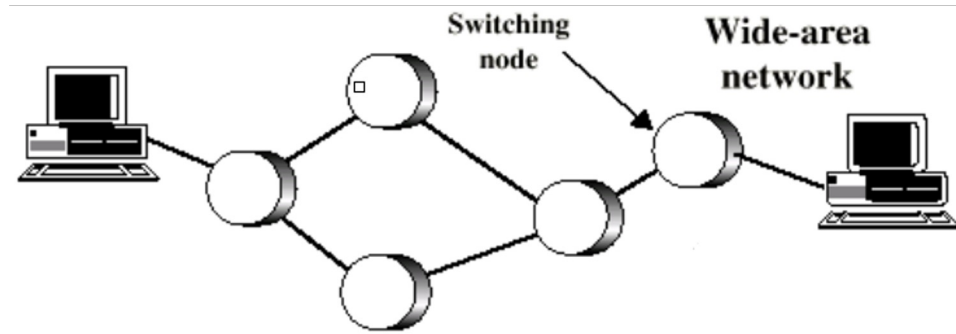




# Wide Area Networks (WANs)

- Characteristics

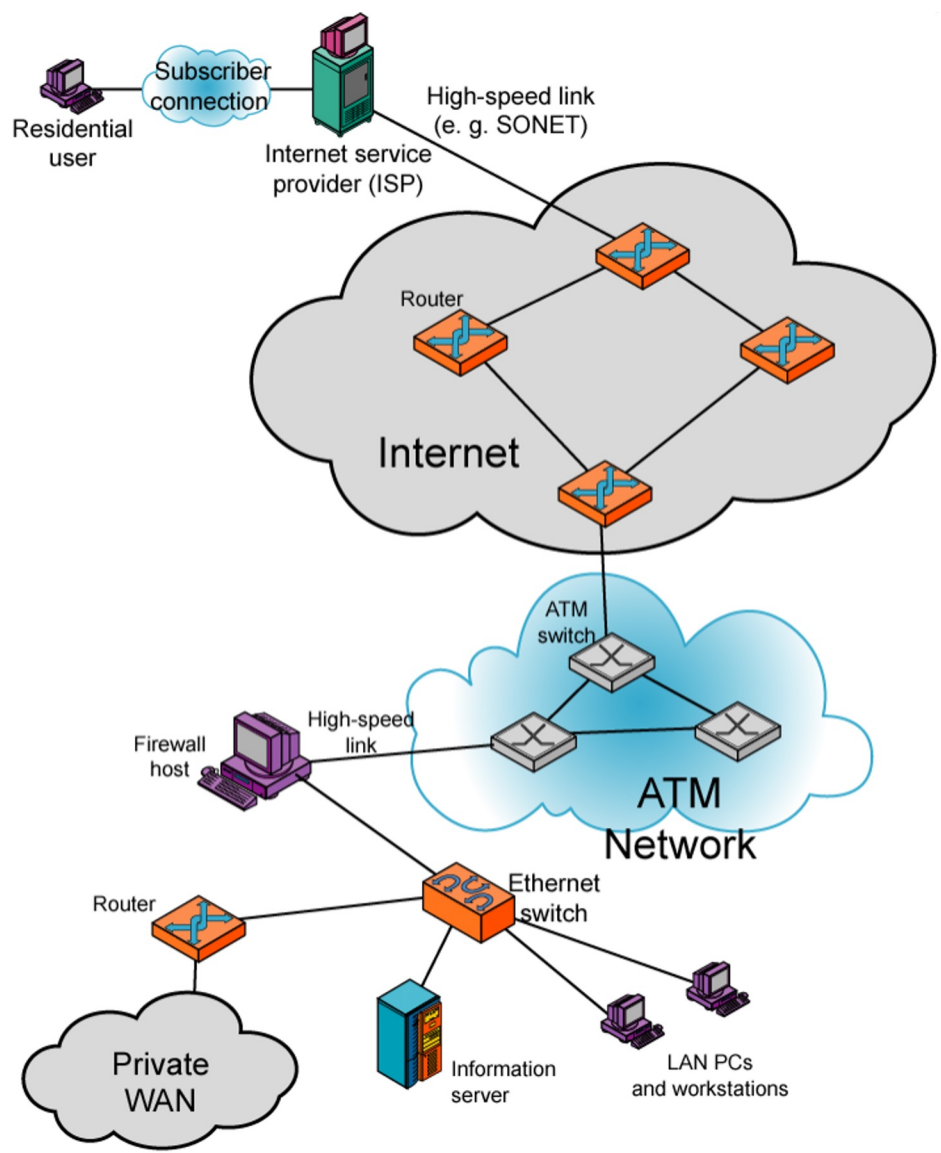
- Span a large geographical area
- Cross public rights of way



- Technologies used include:

- Circuit switching
- Packet switching
- Frame relay
- Asynchronous Transfer Mode (ATM)

# Internet Network



TCP/IP  
Architecture

Application

Transport

Internet

Data Link

Physical

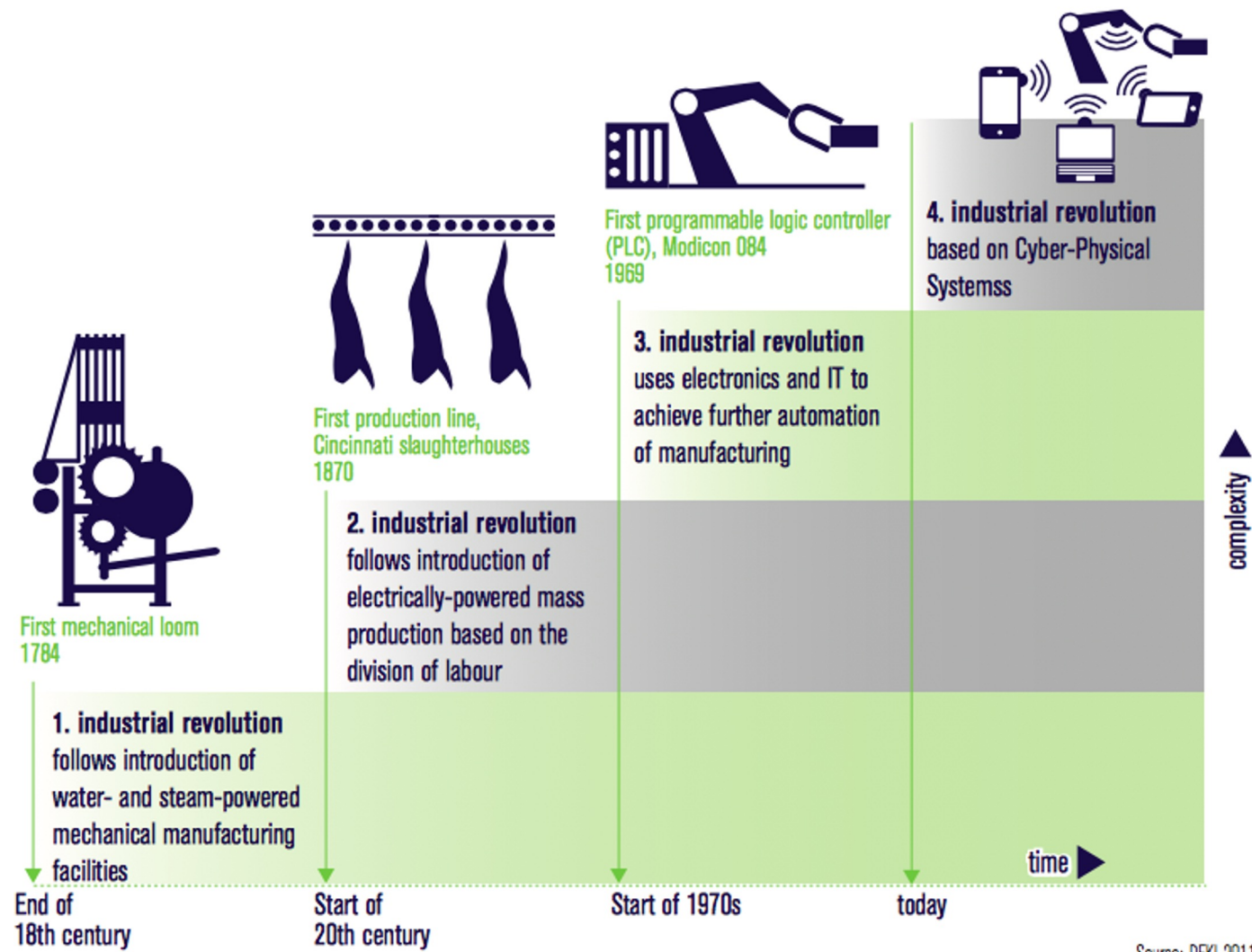


# Internet of Things (IoT)



# Industry 4.0

Figure 1:  
The four stages of  
the Industrial Revolution

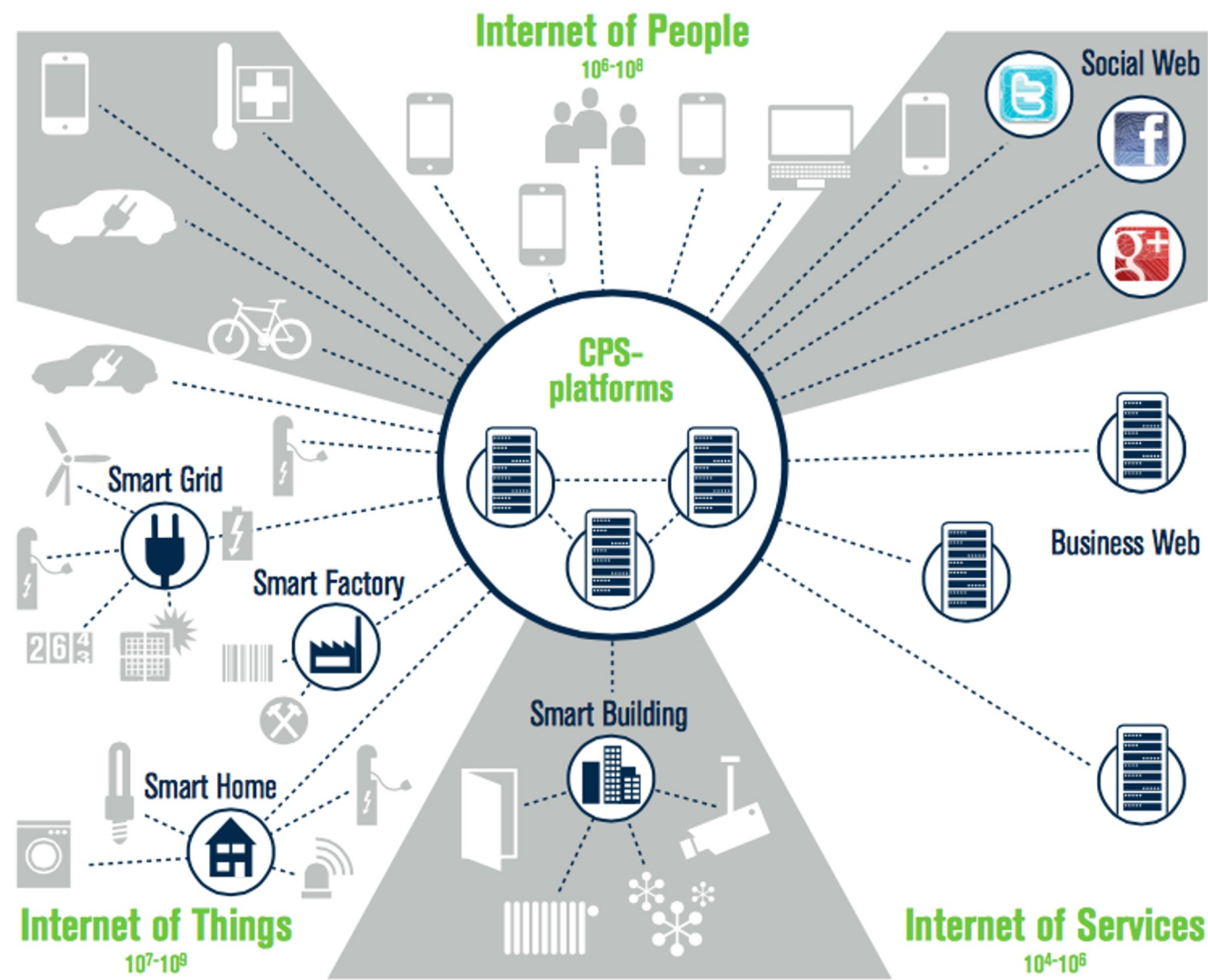


Source: DFKI 2011

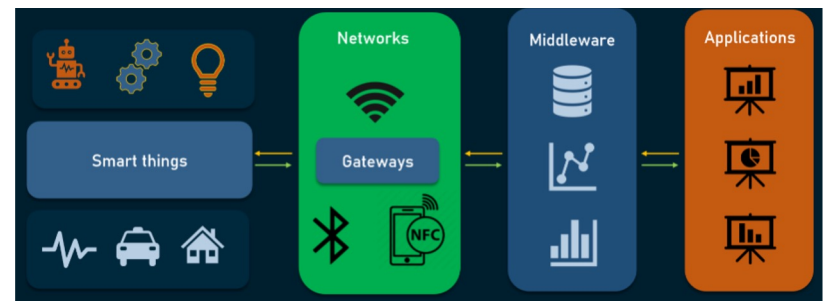
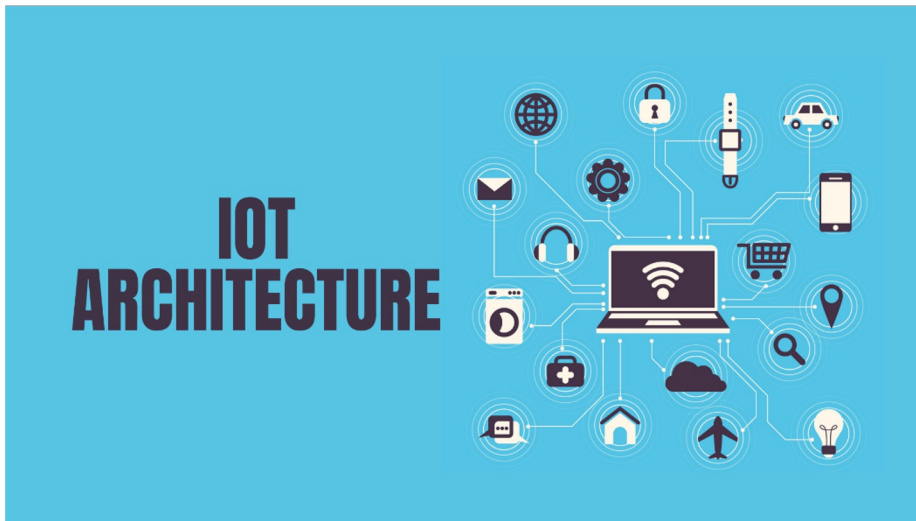


# IoTs and Services

Figure 4:  
The Internet of Things and  
Services – Networking  
people, objects and systems



# Internet of Things Architecture



<https://www.altexsoft.com/>

# Internet of Things (IoT)

(Timothy Chou)

Do

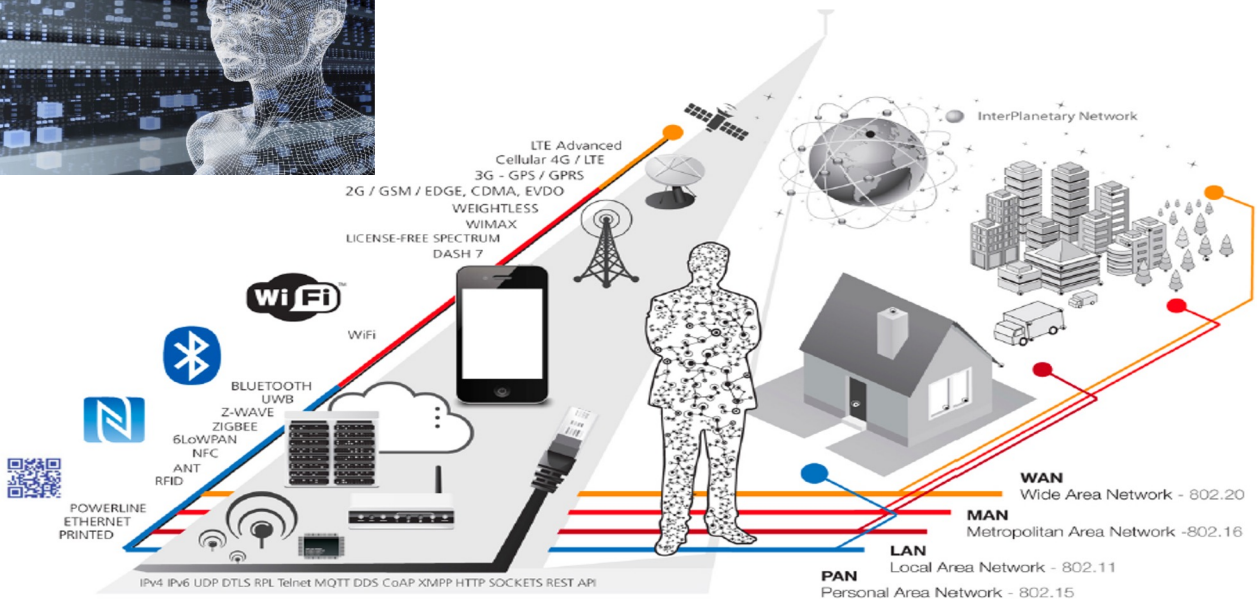


Learn



Collect

Connect



Things



Ambient Light



Touch Screen



Proximity



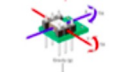
Fingerprint



Attitude



Barometer



Accelerometer



Gyroscope



Moisture



Magnetometer



Gravity



- # Smart Agriculture

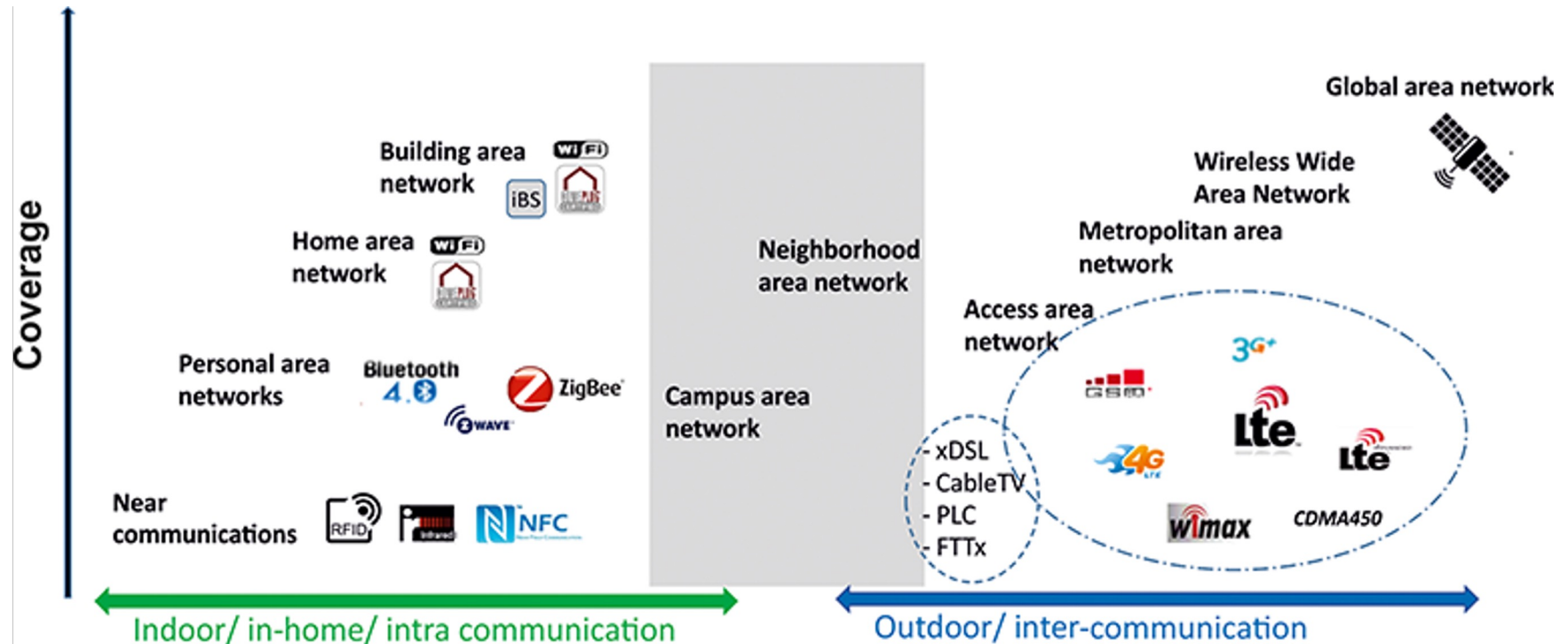


The diagram illustrates the MySIGNAL system architecture. A central white device labeled 'MySIGNAL' is connected to several sensors and monitoring units. The sensors include a Body Position Sensor, Temperature Sensor, EKG Sensor, SpO2 Sensor, CO2 Sensor, Blood Pressure Sensor, Blood Glucose Sensor, and SpO2 Pulse Oximeter. The monitoring units include a SpO2 Pulse Oximeter, a Blood Glucose Monitor, a SpO2 Pulse Oximeter, and a Blood Pressure Monitor. The system is designed to monitor multiple vital signs simultaneously.

## Water Quality



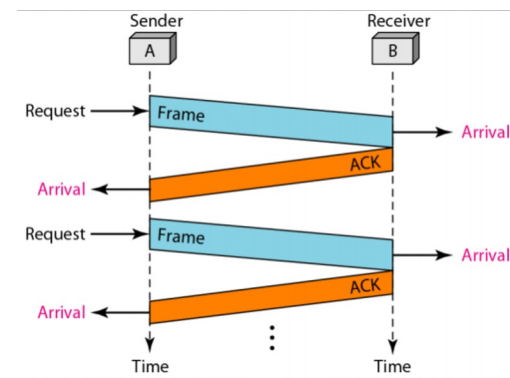
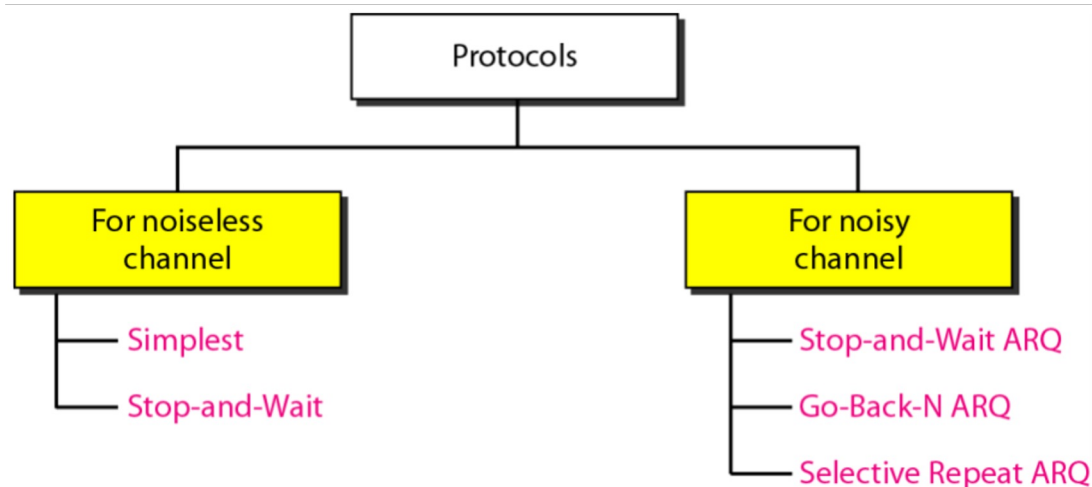
# Connect Layer



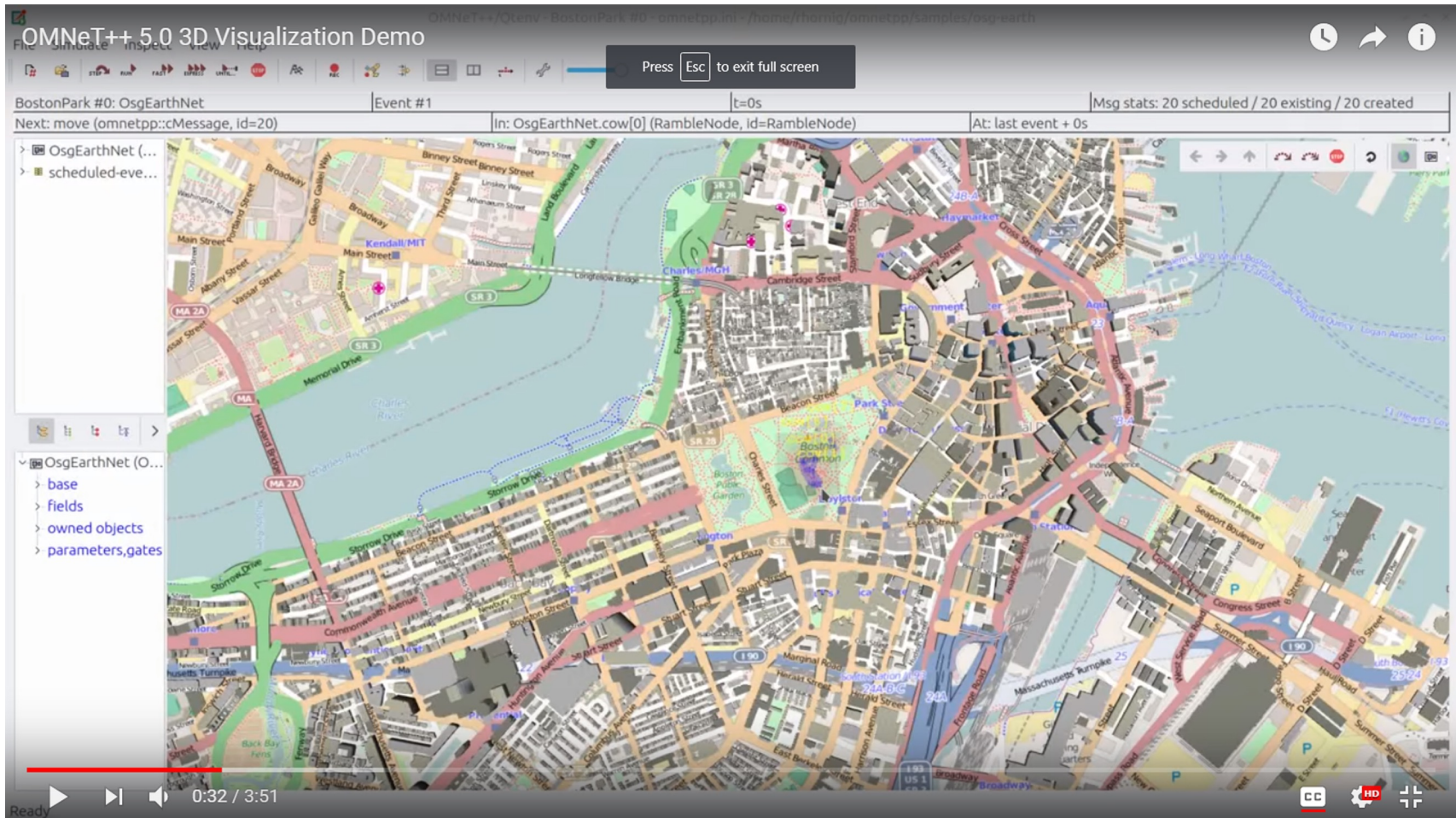
- More than 60 protocols are proposed every year
- Mobility network

# Communication Protocol

- The combination of **framing**, **flow control**, and **error control** to achieve the delivery of data from one node to another.
- The protocols are normally **implemented in software** by using one of the common programming languages.

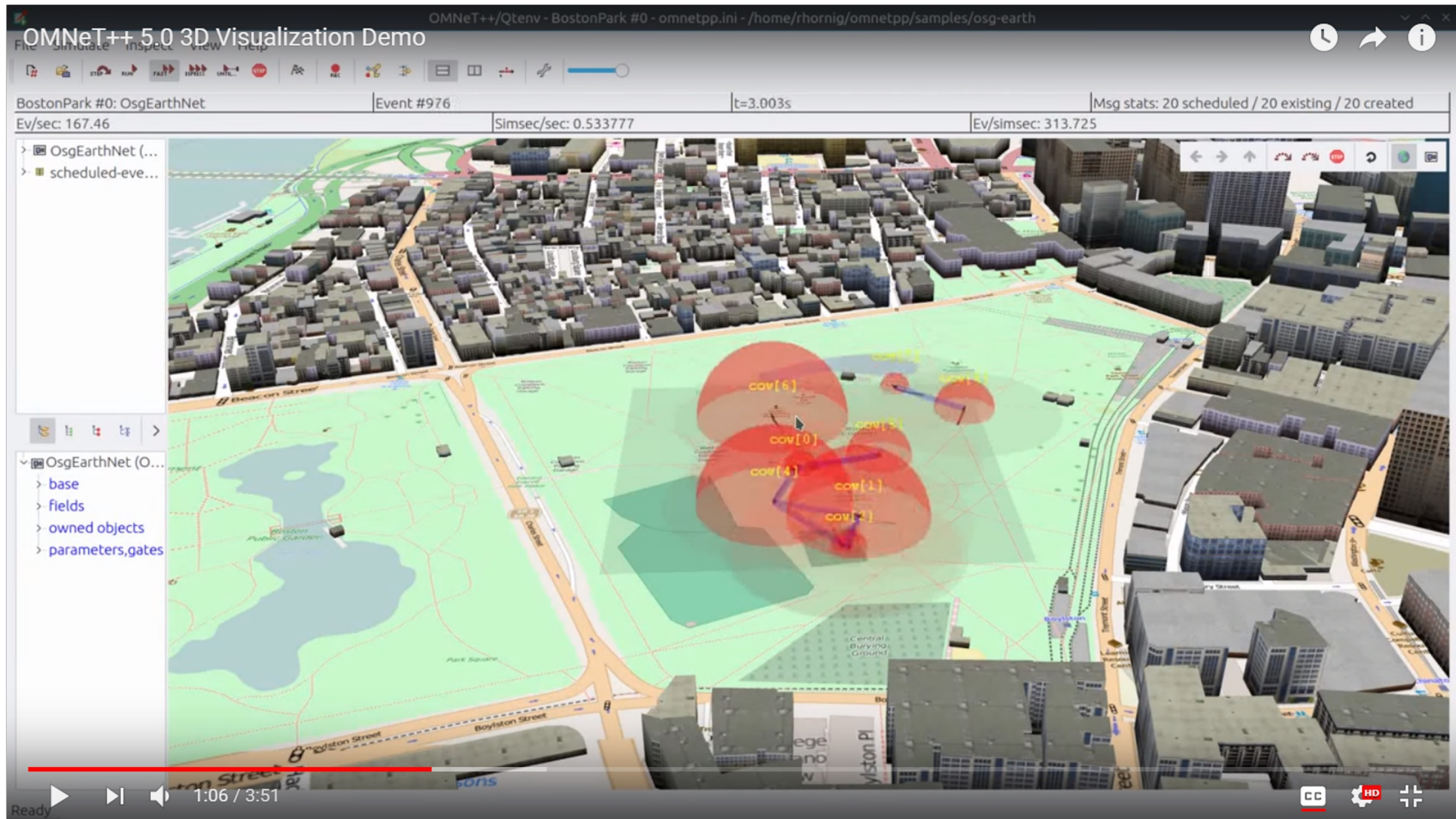


# OMNeT++ 5.0



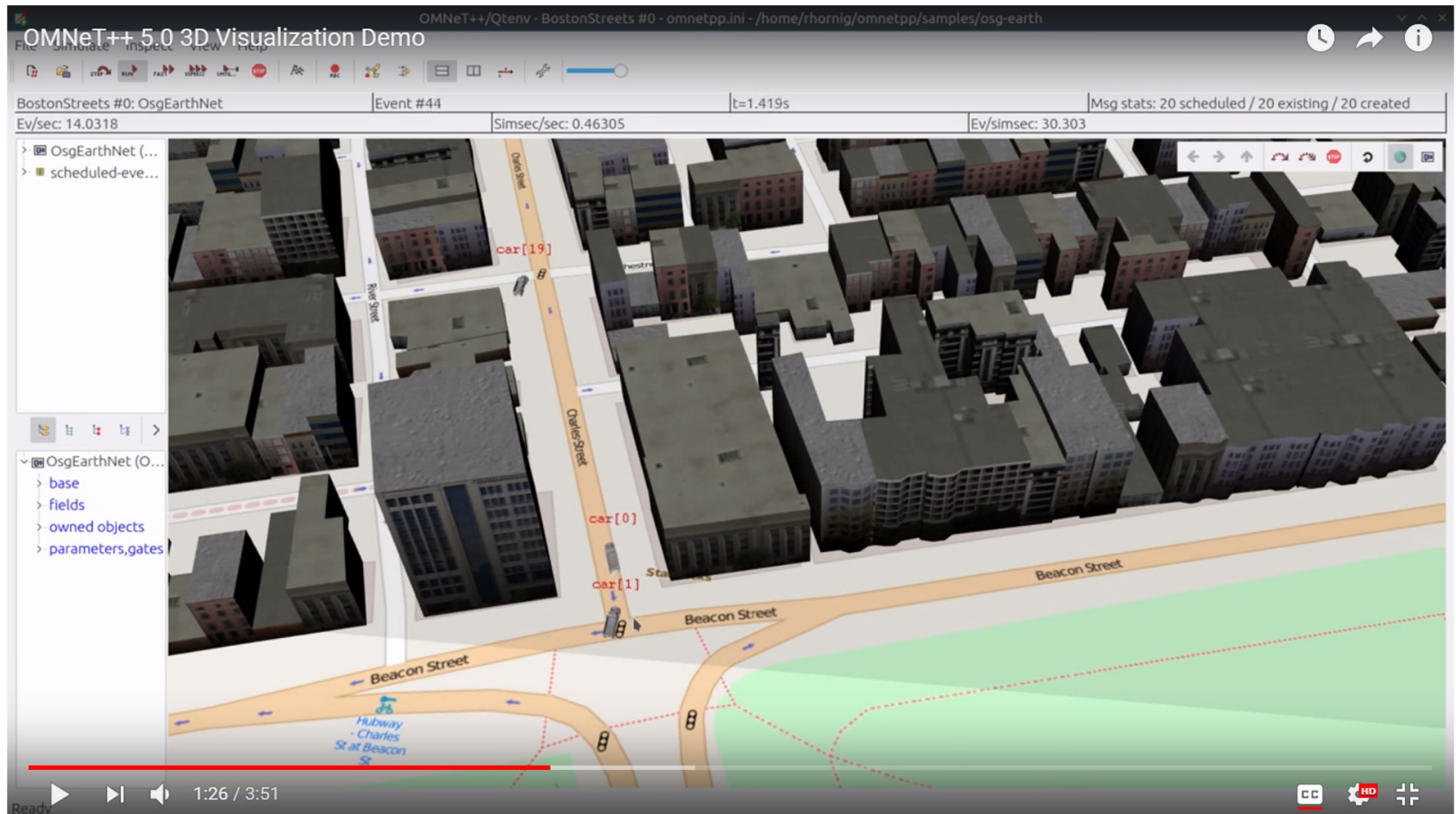


# OMNeT++ 5.0

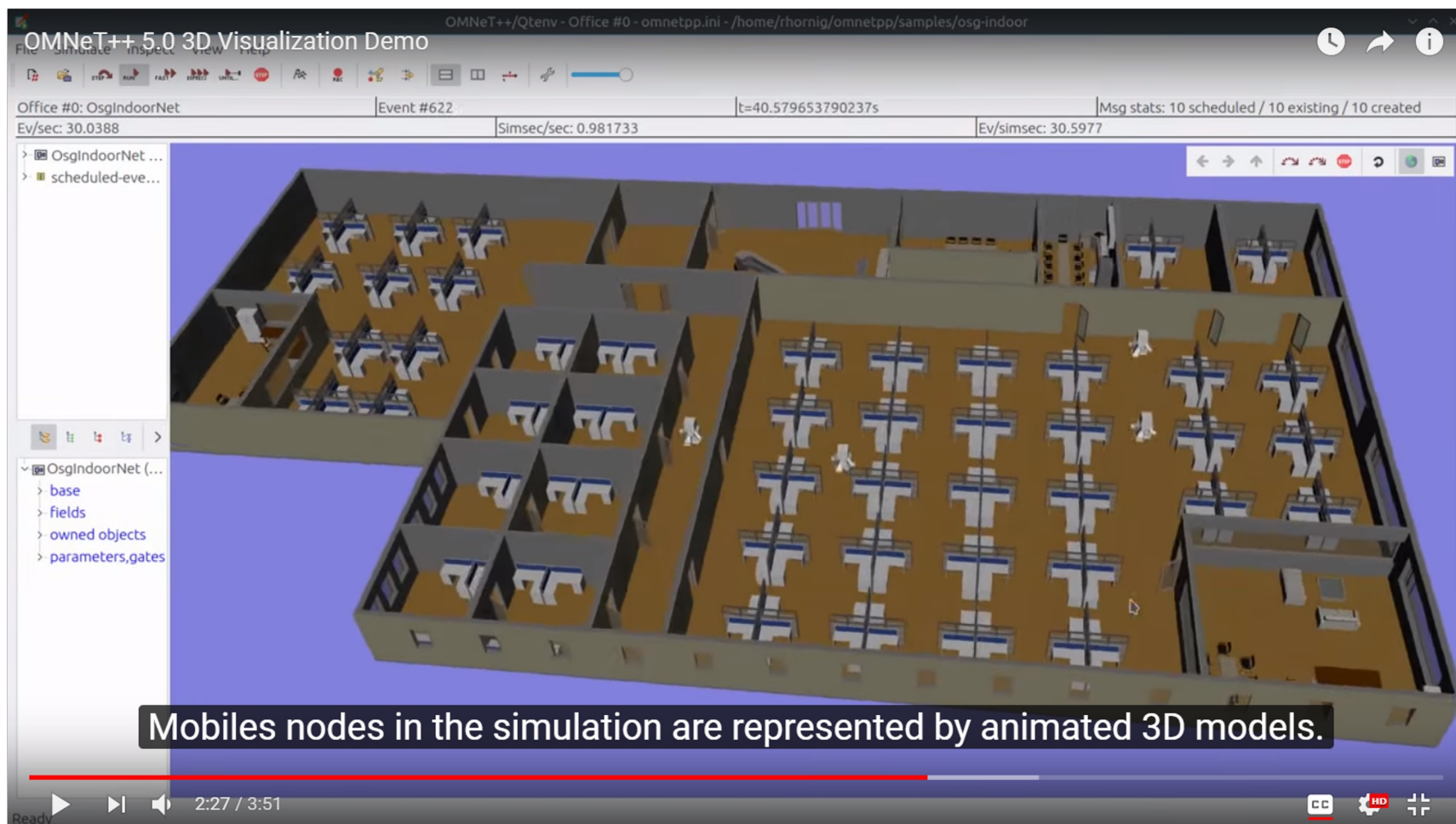




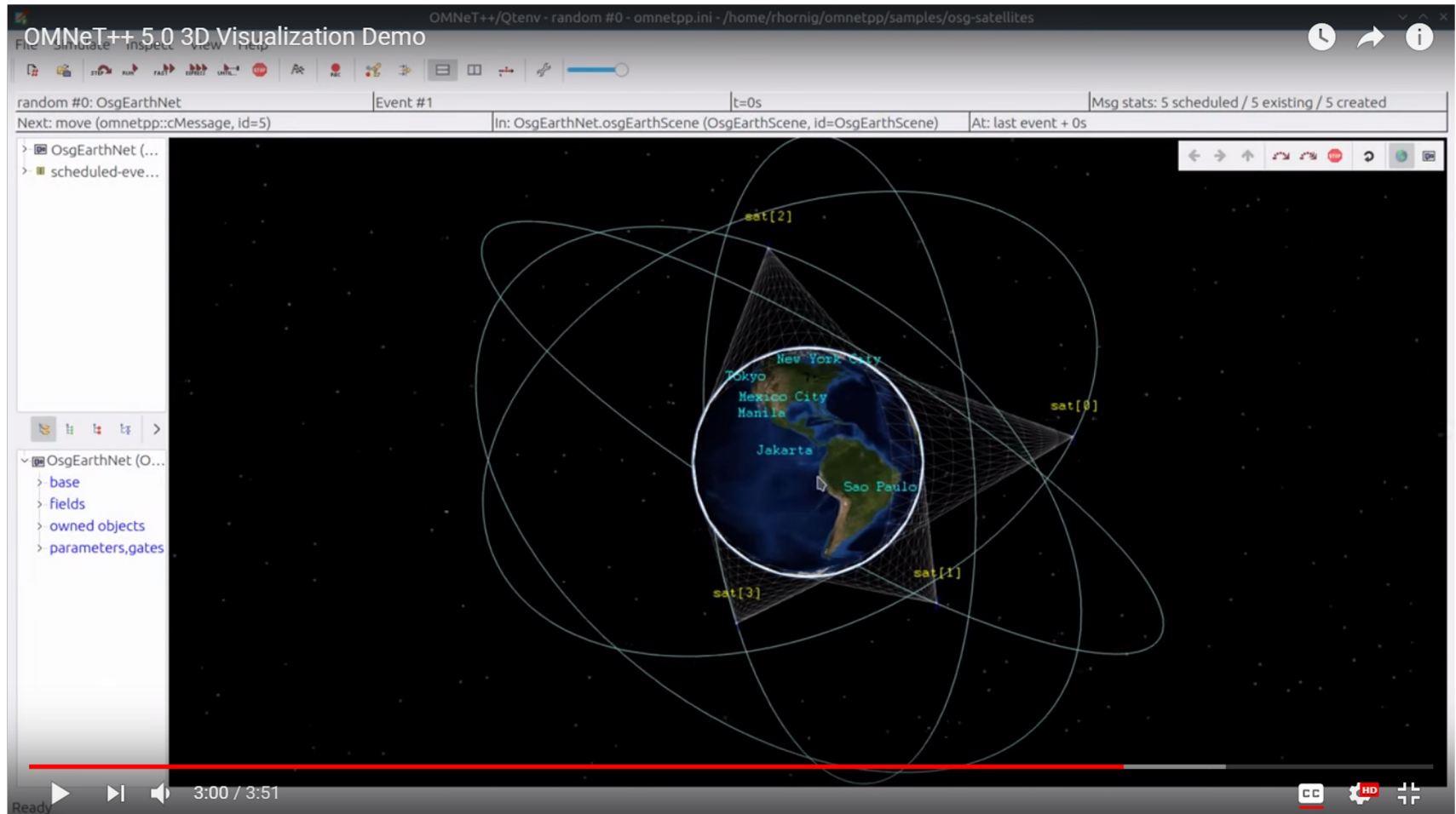
# OMNeT++ 5.0



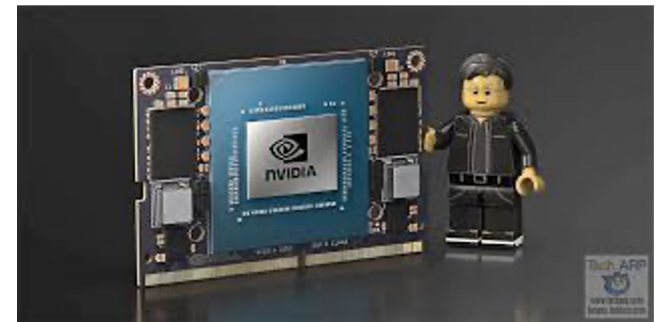
# OMNeT++ 5.0



# OMNeT++ 5.0

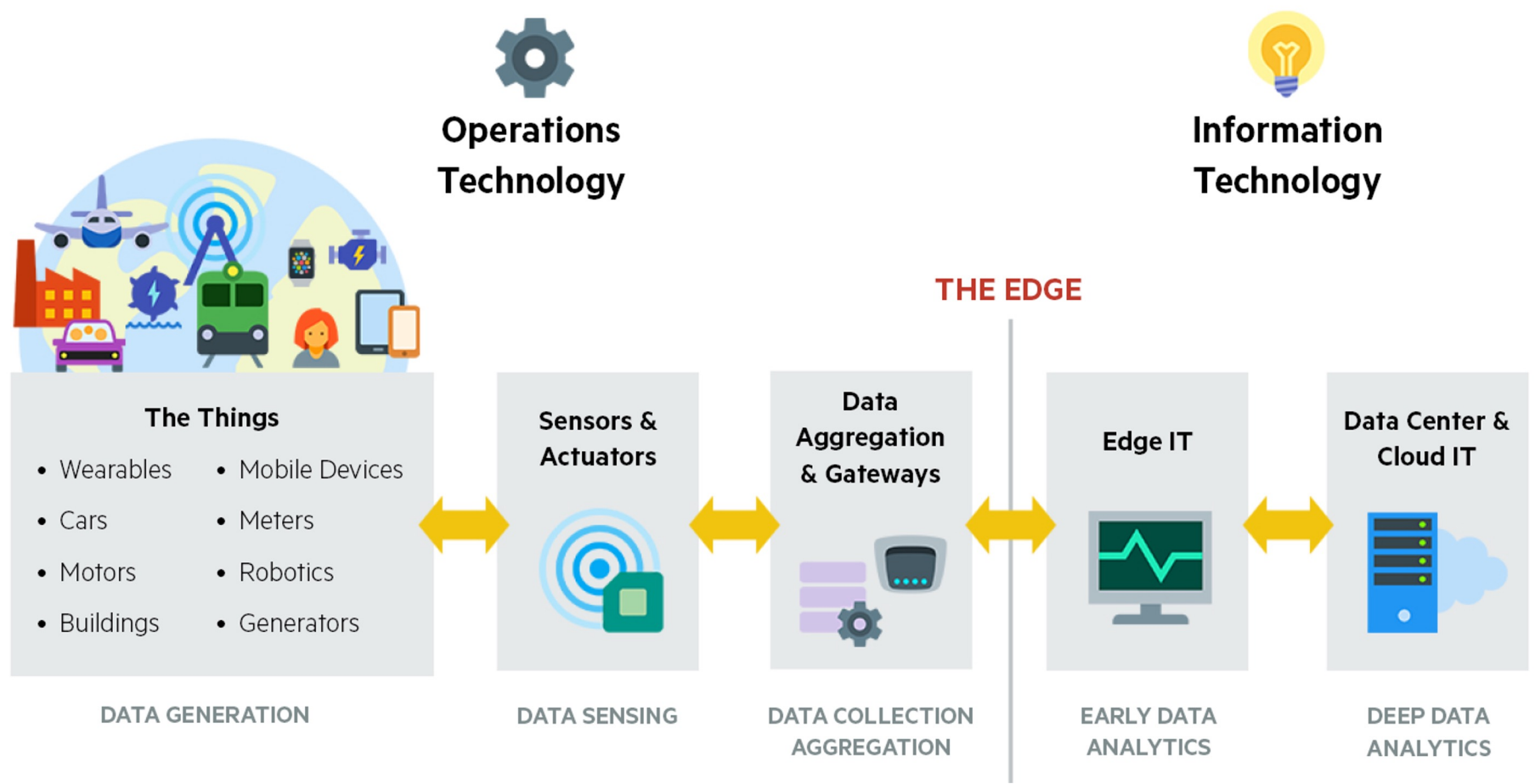


# Edge Computing in Internet of Things Applications



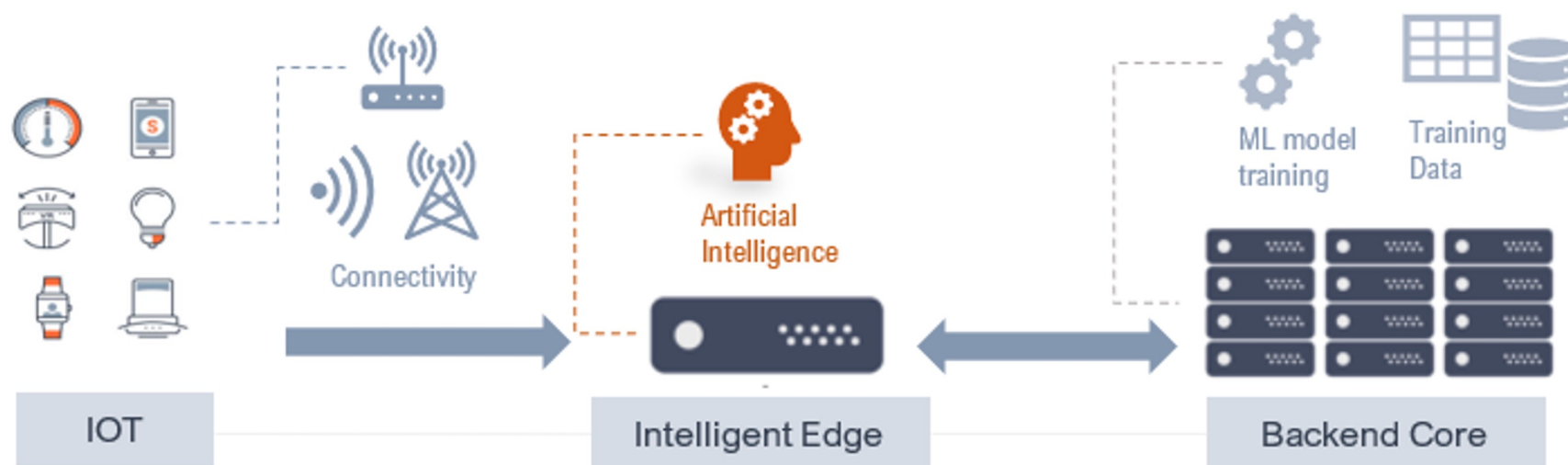


# Edge Computing





# Combine it with AI = Edge AI



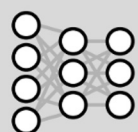
- Edge AI overcomes the latency of data and **complex computations**:
  - Increase in levels of Automation
  - Digital Twins for Advanced Analytics
  - Real-Time Decision Making
  - Edge Inference and Training

# Edge AI on Embedded Platform

Sample Code

Nsight Developer Tools

Multimedia API



TensorRT  
cuDNN  
TF, PyTorch, ...

Deep Learning



VisionWorks  
OpenCV  
NPP

Computer Vision



Vulkan  
OpenGL  
EGL/GLES

Graphics



libargus  
GStreamer  
V4L2

Media

CUDA, Linux For Tegra, ROS

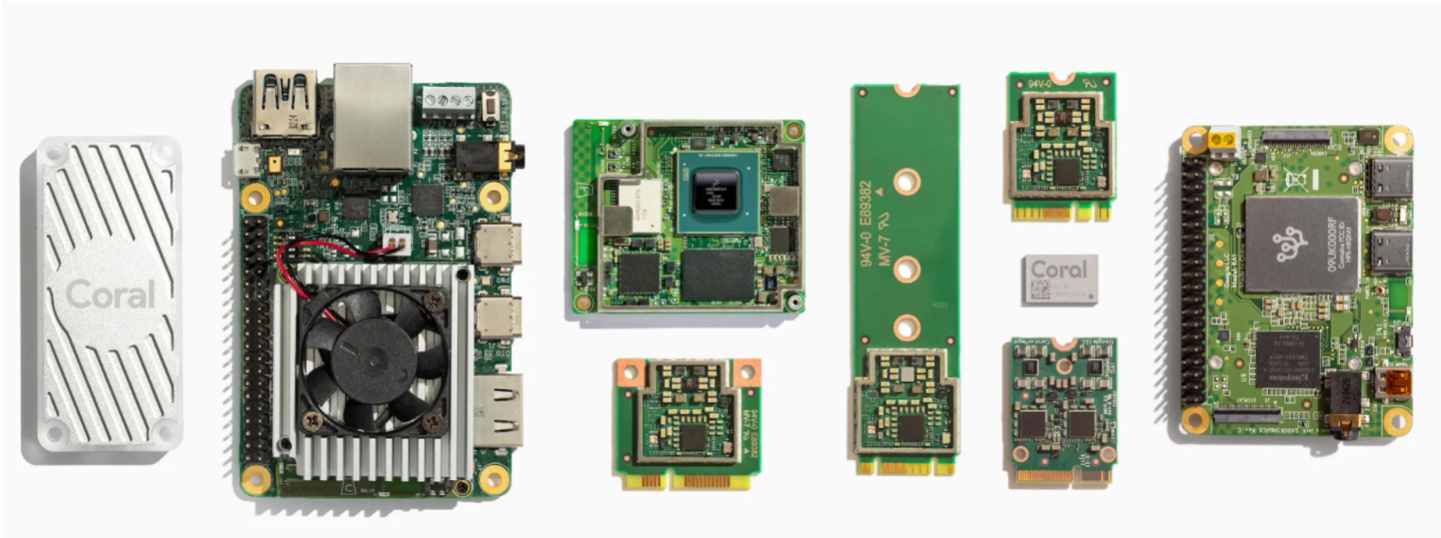


Jetson AGX Xavier: Advanced GPU, 64-bit CPU, Video CODEC, DLAs

384 NVIDIA CUDA cores and 48 Tensor Cores



# PyCoral [<https://coral.ai/products/>]



## Models

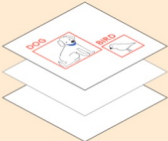
Trained TensorFlow models for the Edge TPU



Image classification

Models that recognize the subject in an image, plus classification models for on-device transfer learning.

→ See models



Object detection

Models that identify multiple objects and provide their location.

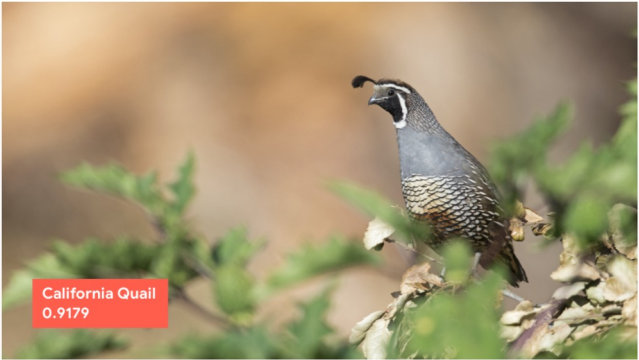
→ See models



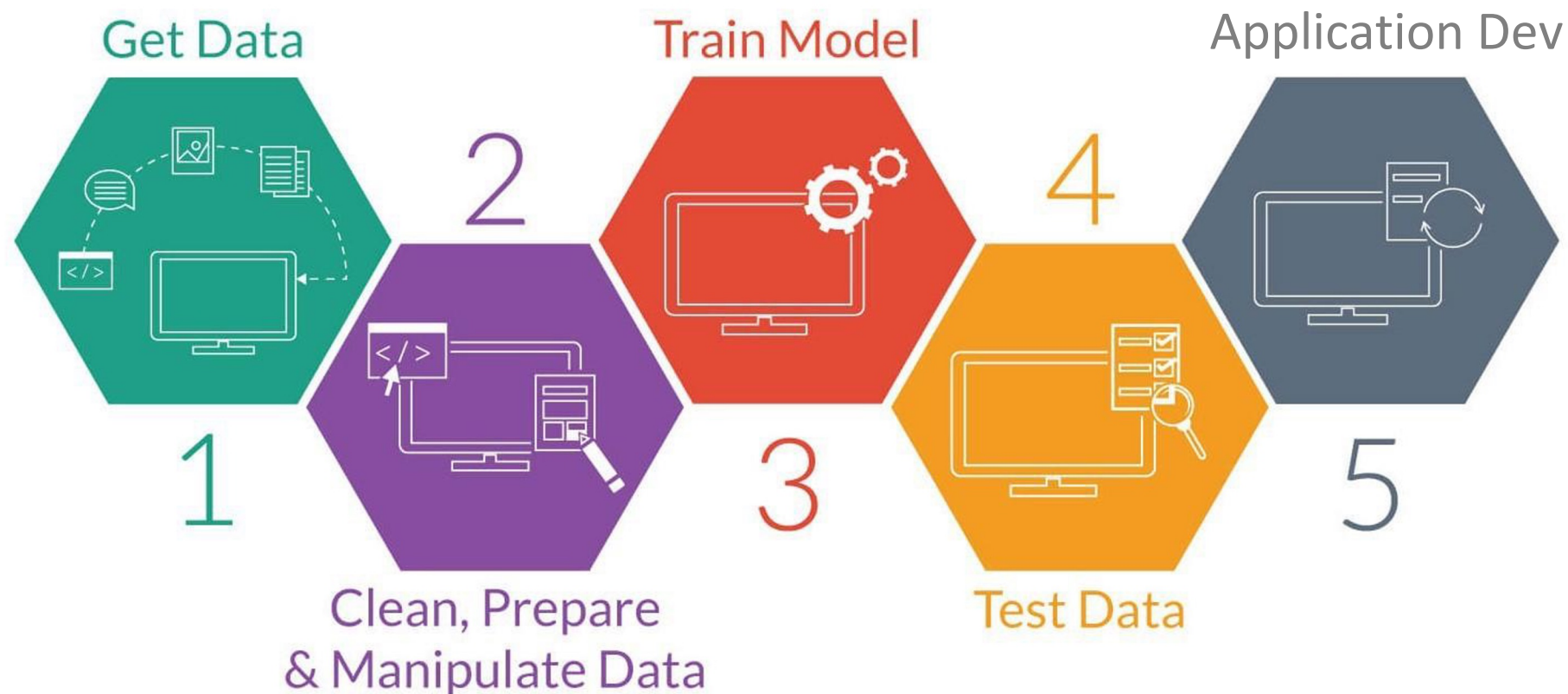
Semantic segmentation

Models that identify specific pixels belonging to different objects.

→ See models



# Train an AI Model



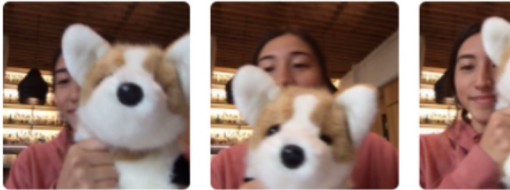
# Demo with Google TM

- <https://teachablemachine.withgoogle.com/train>

## New Project

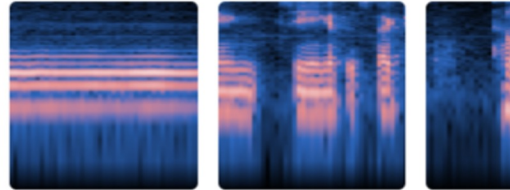
 Open an existing project from Drive.

 Open an existing project from a file.



### Image Project

Teach based on images, from files or your webcam.



### Audio Project

Teach based on one-second-long sounds, from files or your microphone.

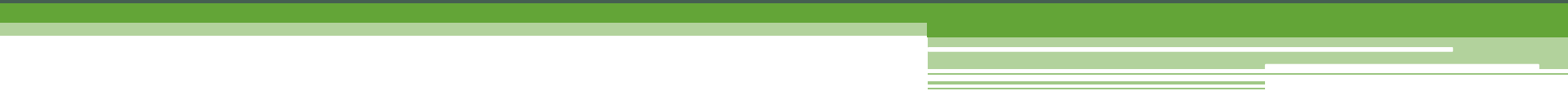


### Pose Project

Teach based on images, from files or your webcam.



# Potential Applications

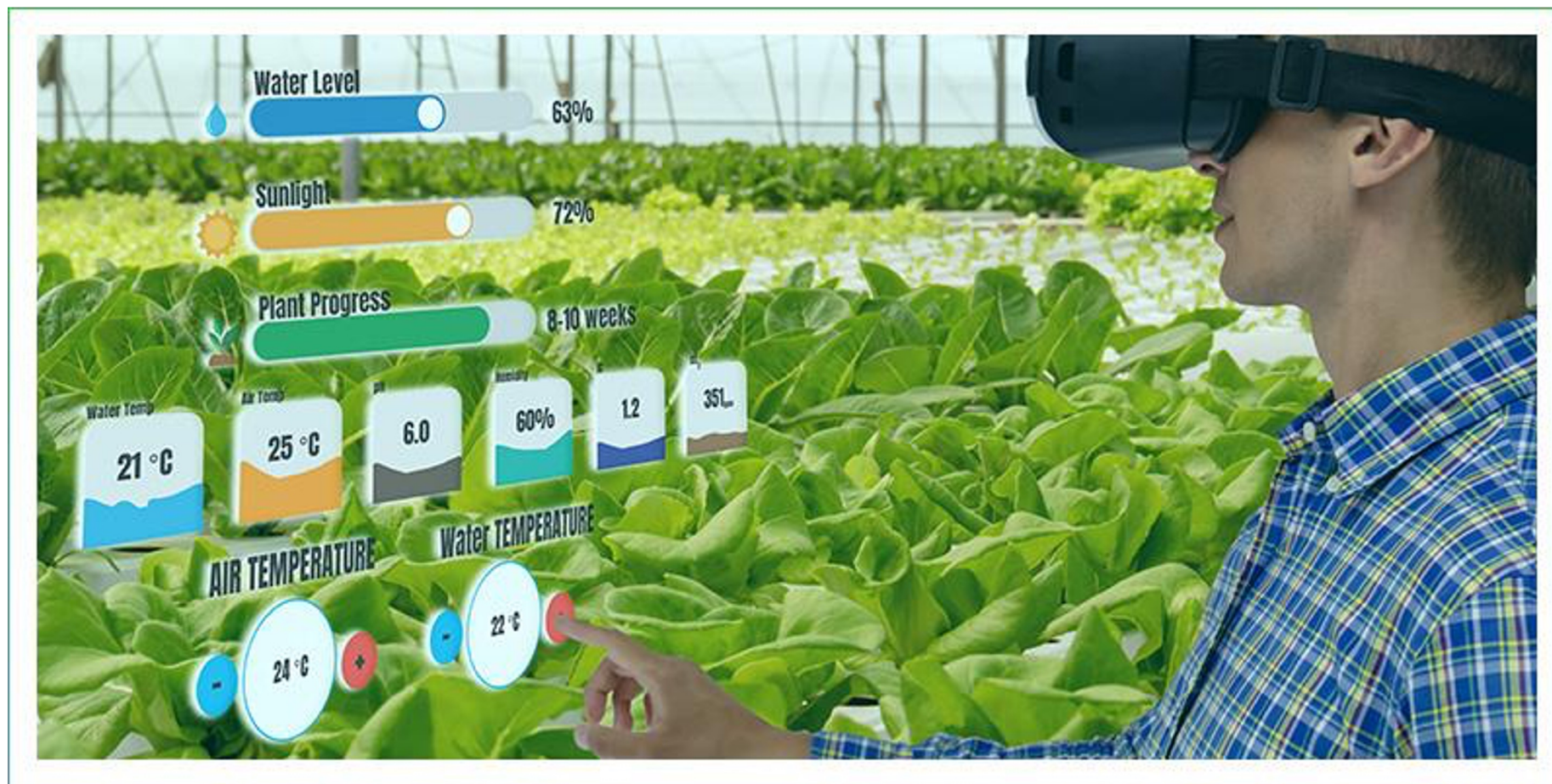
A series of horizontal lines in various shades of green and white, extending across the width of the slide below the title.

# Smart Agriculture

- Temperature and Humidity
- pH, EC
- Light intensity
- Pump controllers

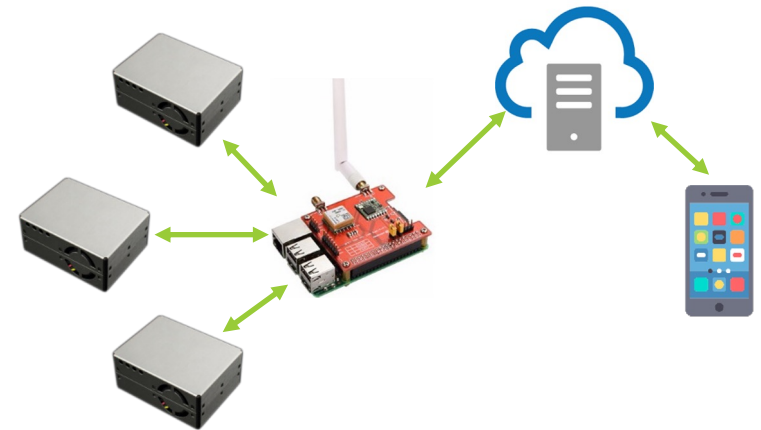
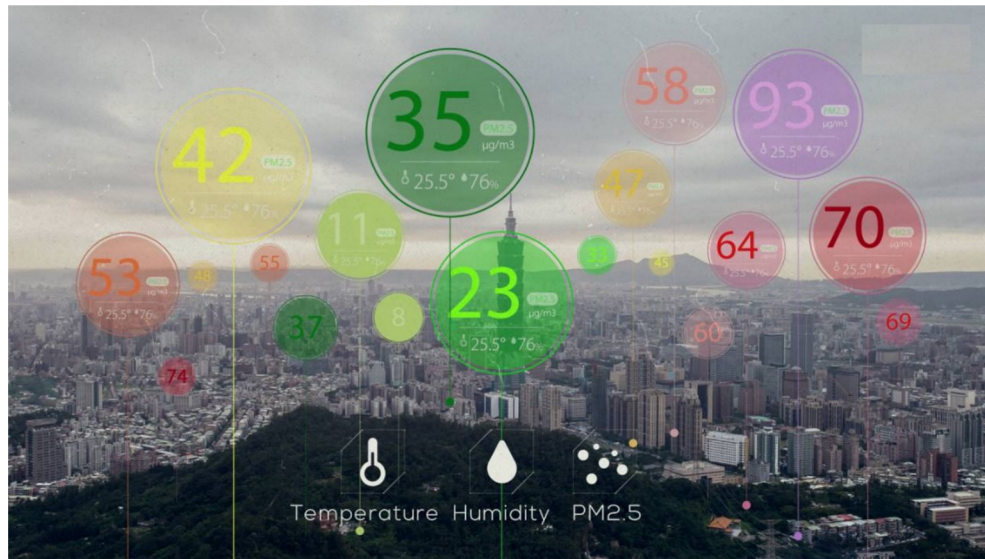


# Monitoring system based AR/VR



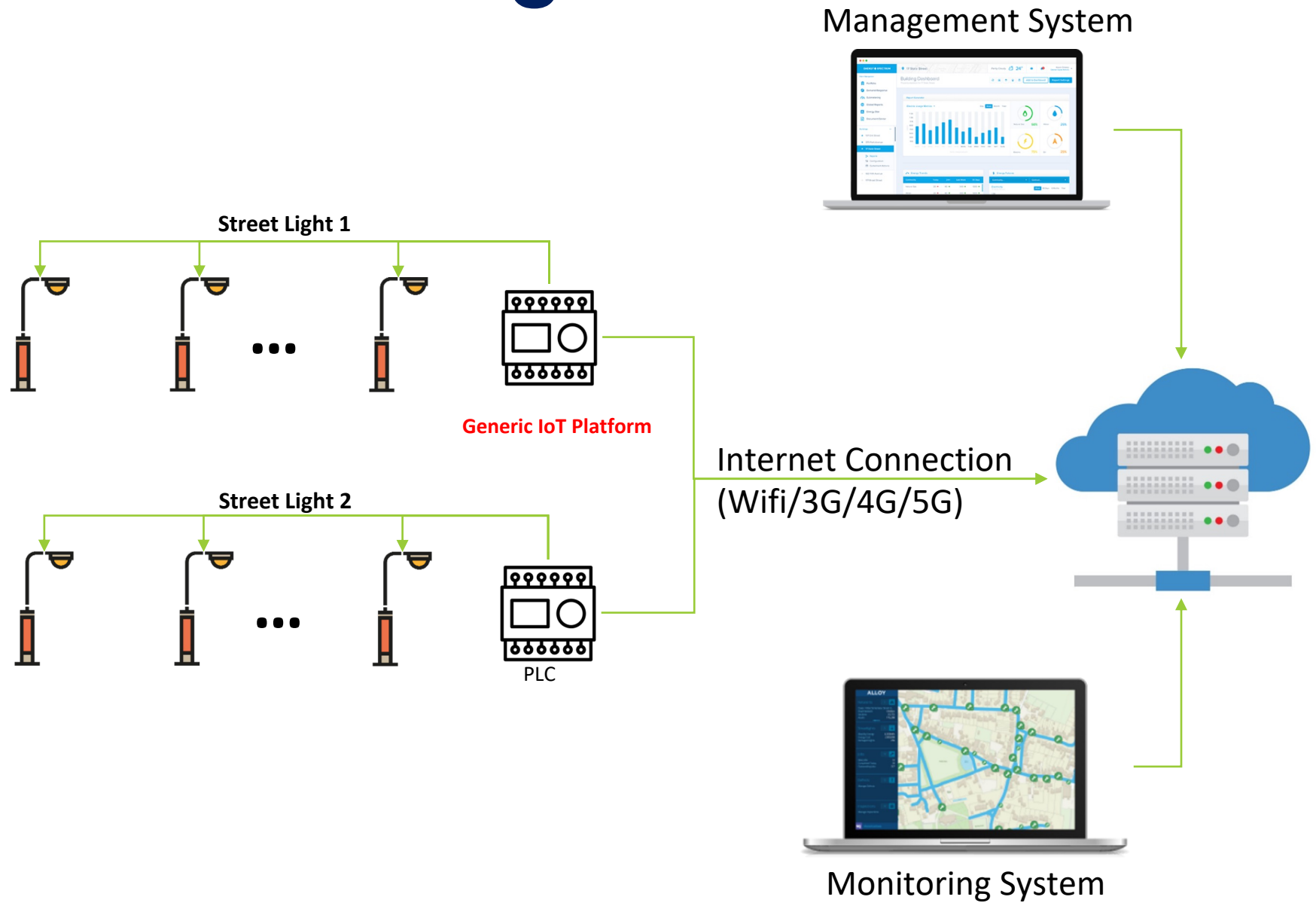


# Air Quality Monitoring



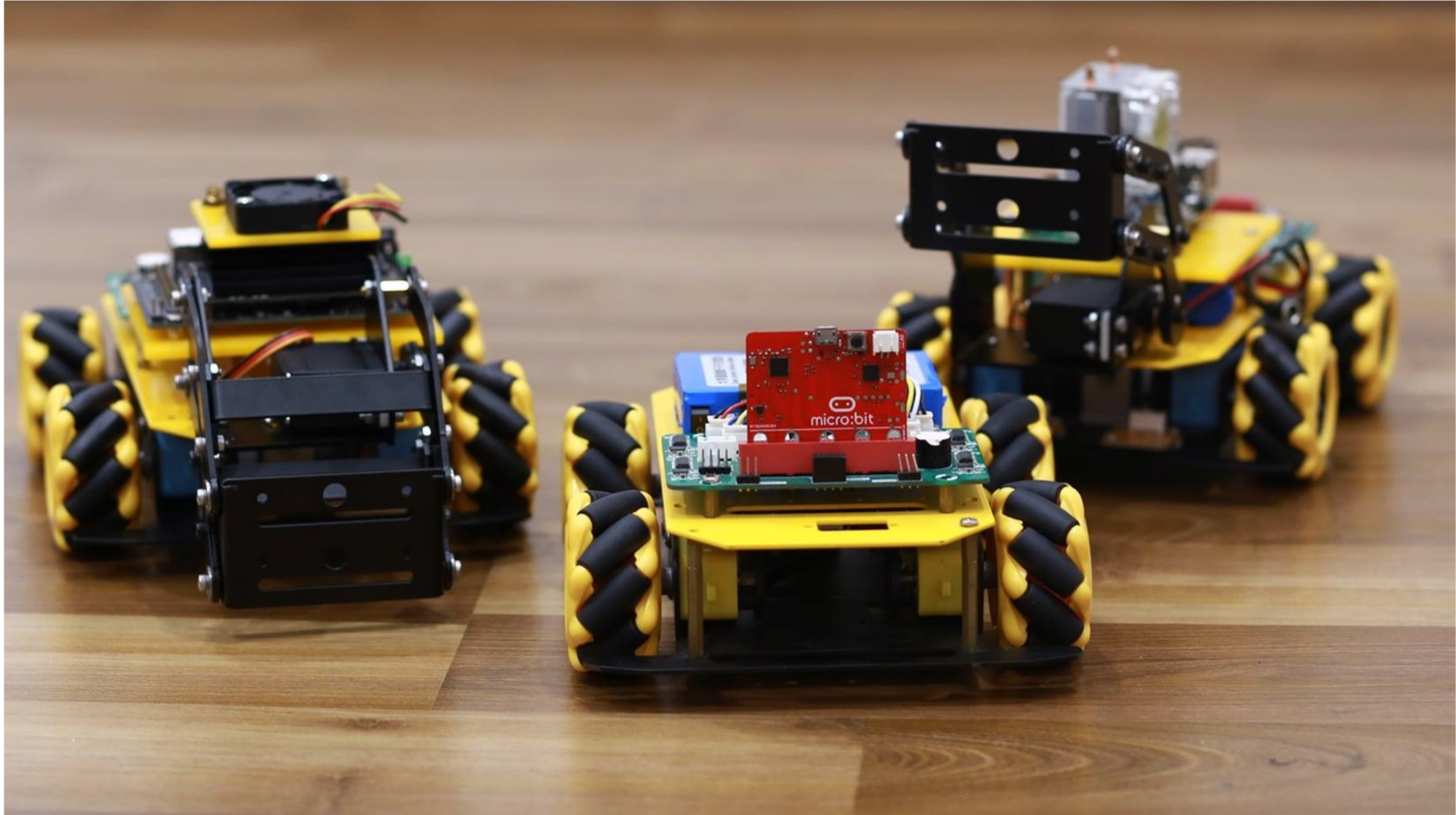
- Air quality monitoring program assists us in improving and **developing air pollution control programs** to reduce the effect of air pollution.
- **PM2.5, PM10, CO2, CO**

# Smart Street Light





# Autonomous Robots



- Smart warehouse applications